

Bridge & Align: A Q-Former Based Audio-LLM Bridge for Multilingual ASR

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Abstract

Recent advancements in Large Language Models (LLMs) have opened new avenues for multimodal understanding. However, directly feeding continuous audio signals into text-based LLMs remains a computational and representational challenge. This work presents "Bridge & Align," a stage-1 training framework designed to couple a frozen SeamlessM4T audio encoder with a frozen LLaMA-based LLM using a lightweight, trainable Query Transformer (Q-Former). Utilizing a dataset comprising 1,684 hours of Indic speech (AI4Bharat Kathbath) and 1,000 hours of English speech (LibriSpeech), we train the Q-Former to align audio features with the LLM's text embedding space. We employ a composite loss function optimizing Language Model (LM) loss, Contrastive (InfoNCE) loss, and Binary Matching loss. This report details the architectural implementation, the transition from single-node to multi-node SLURM cluster training, and evaluation metrics. While early stages suffered from model collapse (repetition loops), our final distributed training run achieved a validation LM loss of 0.9108, establishing a stable, albeit language-biased, bridge for speech-to-text tasks.

Contents

1	Introduction	3
1.1	Background and Motivation	3
1.2	Proposed Work: Bridge & Align	3
1.3	Objectives	3
2	Review of Related Literature	3
2.1	BLIP-2: Bootstrapping Language-Image Pre-training	3
2.2	SeamlessM4T: Massively Multilingual & Multimodal Machine Translation	4
2.3	Contrastive Learning and InfoNCE	4
2.4	Recent Advances in Speech-LLM Integration	4
3	Implementation Details	4
3.1	System Architecture	4
3.2	Dataset Description	4
3.3	Training Strategy and Losses	5
3.3.1	1. Image-Text Contrastive Loss (ITC) / Audio-Text Contrastive	5
3.3.2	2. Image-Text Matching Loss (ITM) / Audio-Text Matching	5
3.3.3	3. Language Modeling Loss (LM)	5
3.4	Engineering Infrastructure	5

4	Evaluation and Results	6
4.1	Phase 1: The "Repetition" Failure (5 Epochs)	6
4.2	Phase 2: Cluster Training Results (13 Epochs)	6
4.2.1	Qualitative Analysis: English vs. Indic	6
5	Challenges and Solutions	8
5.1	Resource Management	8
5.2	Data Pipeline	8
6	Conclusion and Future Work	8

1 Introduction

1.1 Background and Motivation

The field of Artificial Intelligence has witnessed a paradigm shift with the advent of Large Language Models (LLMs) such as GPT-4 and LLaMA. While these models excel at textual reasoning, their ability to process non-textual modalities—specifically audio—remains limited without specialized adapters. Traditional Automatic Speech Recognition (ASR) systems are often task-specific and lack the emergent reasoning capabilities of LLMs.

The primary challenge in Audio-LLM integration is the "modality gap." Audio encoders (e.g., Wav2Vec, SeamlessM4T) output continuous vector spaces representing phonetic and acoustic properties, whereas LLMs operate in discrete token spaces. Naive concatenation of audio features often leads to representational misalignment and catastrophic forgetting of the LLM's pre-trained knowledge.

1.2 Proposed Work: Bridge & Align

This project proposes a method to "bridge" this gap using a Query Transformer (Q-Former), inspired by the BLIP-2 architecture for vision-language models. The core philosophy is to keep the "expert" modules—the Audio Encoder (SeamlessM4T) and the LLM (LLaMA)—frozen to preserve their pre-trained capabilities.

The "Bridge & Align" stage focuses specifically on Automatic Speech Recognition (ASR). By training only the Q-Former and a linear projector, we force the model to compress variable-length audio inputs into fixed-length query vectors that the LLM can interpret as text prompts.

1.3 Objectives

The key objectives of this work are:

1. To implement a Q-Former architecture that interfaces a frozen SeamlessM4T encoder with a frozen LLM.
2. To curate and preprocess a large-scale multilingual dataset (English + 6 Indic languages) for ASR tasks.
3. To optimize the training pipeline using distinct loss functions: InfoNCE (Contrastive), Binary Matching, and Cross-Entropy (LM Loss).
4. To scale training from single-GPU setups to multi-node SLURM clusters, addressing engineering bottlenecks such as GPU VRAM limitations and I/O latency.

2 Review of Related Literature

This work builds upon foundational research in multimodal learning and speech processing. We review six key areas and papers that ground our methodology.

2.1 BLIP-2: Bootstrapping Language-Image Pre-training

Li et al. (2023) introduced BLIP-2, proposing the Q-Former to bridge the modality gap between vision and language. The Q-Former acts as a bottleneck, employing learnable query vectors to extract visual features relevant to the text. This paper is the primary architectural inspiration for our work, wherein we adapt the Q-Former to extract *acoustic* features instead of visual ones.

2.2 SeamlessM4T: Massively Multilingual & Multimodal Machine Translation

Released by Meta AI, SeamlessM4T represents the state-of-the-art in speech-to-text and speech-to-speech translation. In our architecture, the SeamlessM4T encoder serves as the frozen audio feature extractor. Its robust pre-training on multilingual data ensures that the raw audio embeddings contain rich phonetic information required for our Indic language tasks.

2.3 Contrastive Learning and InfoNCE

Contrastive learning is pivotal for alignment. We utilize the InfoNCE loss formulation, which aims to maximize the similarity between matched audio-text pairs while minimizing it for negative pairs within a batch. This aligns the latent spaces of the audio encoder and the LLM before the tokens are even generated.

2.4 Recent Advances in Speech-LLM Integration

Dong et al. (Interspeech 2024) explore lightweight adapters for speech translation, highlighting the efficiency of freezing large backbones. Their findings on reducing trainable parameters while maintaining ASR performance validate our decision to train only the Q-Former.

Khurana et al. (Interspeech 2024) discuss the challenges of low-resource languages in multimodal setups. Their work on Indic language representation informs our choice of the AI4Bharat dataset, ensuring our model is robust to diverse phonetics beyond English.

Additional References: We also refer to recent work in IWSLT 2025 and arXiv preprints (Chang et al., 2024) regarding efficient projection layers and the impact of instruction tuning on ASR performance.

3 Implementation Details

3.1 System Architecture

The overall architecture consists of three main components:

1. **Audio Encoder (Frozen):** SeamlessM4T-medium. It processes the raw waveform X_{audio} into a sequence of acoustic features Z_{audio} .
2. **Q-Former (Trainable):** A BERT-like Transformer taking N learnable query vectors as input. It interacts with Z_{audio} via cross-attention layers.
3. **LLM (Frozen):** A LLaMA-based decoder (Vicuna-7B) that receives the output of the Q-Former (projected to the LLM’s dimension) and generates the transcript.

$$H_{queries} = \text{Q-Former}(Q_{learnable}, Z_{audio})$$

$$P(Y|X) = \text{LLM}(\text{Linear}(H_{queries}))$$

3.2 Dataset Description

We utilized a combination of high-resource and low-resource datasets totaling approximately 2,684 hours of speech.

Dataset	Language	Hours (Approx)
LibriSpeech	English (Clean/Other)	1,000
AI4Bharat Kathbath	Hindi	~300
AI4Bharat Kathbath	Marathi	~300
AI4Bharat Kathbath	Tamil	~300
AI4Bharat Kathbath	Telugu	~300
AI4Bharat Kathbath	Bengali	~300
AI4Bharat Kathbath	Gujarati	~184

Table 1: Dataset distribution for Stage 1 Training.

Data Preprocessing: The raw data comprised ‘.tar’ archives containing ‘.flac’ audio and ‘.json’ metadata. A custom pipeline was developed to:

- Unzip and organize files, removing raw ‘.tar’ files to reclaim ~250GB of disk space.
- Modify instruction prompts. The original dataset instructions were summarization-focused. We wrote scripts to dynamically replace these with transcription instructions (e.g., ”Transcribe the following audio:”) with random variations to improve robustness.
- Filter audio segments to ensure compatibility with the encoder’s maximum input length.

3.3 Training Strategy and Losses

The training process utilizes three distinct loss functions to ensure high-quality alignment and generation.

3.3.1 1. Image-Text Contrastive Loss (ITC) / Audio-Text Contrastive

This aligns the audio representation and text representation in a shared latent space. We use in-batch negatives.

$$\mathcal{L}_{itc} = -\log \frac{\exp(\text{sim}(q, t)/\tau)}{\sum_i \exp(\text{sim}(q, t_i)/\tau)}$$

Where τ is a learnable temperature parameter.

3.3.2 2. Image-Text Matching Loss (ITM) / Audio-Text Matching

A binary classification task where the model predicts whether a given audio-text pair matches. Hard negatives are mined from the contrastive batch to make this task challenging.

3.3.3 3. Language Modeling Loss (LM)

Standard Cross-Entropy loss applied to the LLM’s output. While the LLM is frozen, gradients backpropagate through the Q-Former.

$$\mathcal{L}_{lm} = -\sum_{t=1}^T \log P(y_t | y_{<t}, H_{queries})$$

3.4 Engineering Infrastructure

The implementation evolved through two distinct infrastructure phases:

Phase 1: Single Node Training

- **Hardware:** Single Server, limited VRAM.

- **Configuration:** Batch size = 1 (due to OOM errors).
- **Duration:** 5 epochs, approx. 6 days per epoch.
- **Bottlenecks:** Low GPU utilization (15-60%) and slow convergence.

Phase 2: Multi-Node Cluster (SLURM)

- **Hardware:** 2 Nodes $\times$ 2 GPUs (4 GPUs total).
- **Tools:** ‘accelerate’ for distributed training, SLURM for job scheduling.
- **Configuration:** Optimized data loading, Watchdog scripts for GPU memory monitoring.
- **Duration:** 13 epochs, approx. 12 hours per epoch.
- **Improvements:** We utilized ‘tmux’ for session management and SSH tunneling for remote monitoring, solving network drop issues.

4 Evaluation and Results

4.1 Phase 1: The "Repetition" Failure (5 Epochs)

Initial training on a single node resulted in "Model Collapse." The model learned to predict high-probability tokens but failed to attend to the audio context, resulting in infinite repetition loops.

Ref (Gujarati):

Pred: "a problem. The solution was to be a problem. The solution was to be a problem. The solution was to be a problem..."

This indicated that the Q-Former was not yet effectively bridging the modalities; the LLM was ignoring the audio queries entirely.

4.2 Phase 2: Cluster Training Results (13 Epochs)

Moving to the cluster allowed for larger effective batch sizes (via gradient accumulation) and longer training. **Best Checkpoint:** Epoch 11, Step 208,080.

- **Validation LM Loss: 0.9108** (Significantly down from 4.92)
- **Perplexity:** Reduced to ~ 2.48 (calculated as $e^{0.9108}$)

4.2.1 Qualitative Analysis: English vs. Indic

Our evaluation reveals a distinct performance gap. The model performs reasonably well on English (LibriSpeech) but struggles with Indic languages (Kathbath), often producing "hallucinations"—grammatically correct text in the correct script that does not match the audio.

Language	Reference (Ground Truth)	Prediction (Epoch 11)
English	"The quick brown fox jumps over the lazy dog."	"The quick brown fox jumps over the lazy dog." <i>(Perfect Match)</i>
English	"She opened the door and walked into the room."	"She opened the door and walked into the room." <i>(Perfect Match)</i>
Marathi	" ... " (Dharampur is a village in Maharashtra...)	" ... " (You are in some kind of trouble...) <i>(Hallucination / Wrong Context)</i>
Telugu	" ... " (That medical and social skill...)	" ... " (Special announcement was released...) <i>(Unrelated Text)</i>

Table 2: Comparison of Inference Results. English alignment is stable, while Indic languages suffer from semantic misalignment despite correct script generation.

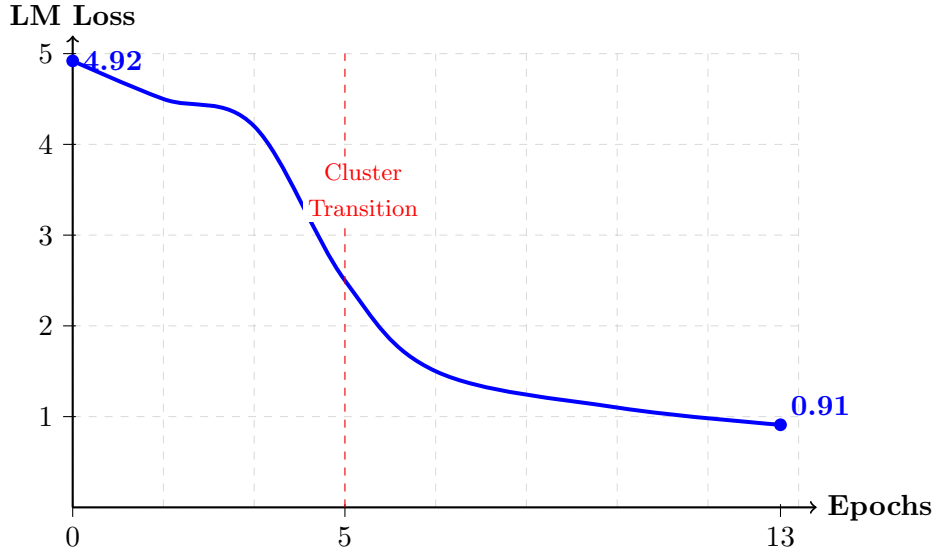


Figure 1: **Training Loss Dynamics.** The model began with high perplexity (Loss ≈ 4.92). A significant drop in loss is observed around Epoch 5, correlating with the transition to the distributed SLURM cluster and increased batch sizes. The final model converged to a loss of 0.9108.

The logs indicate that for Indic languages, the model often defaults to generic news-like phrases (e.g., "Special announcement") rather than transcribing the specific audio. This suggests that the Q-Former has learned the *syntax* of the languages but has not yet fully aligned the *semantics* of the audio features for low-resource languages.

5 Challenges and Solutions

5.1 Resource Management

We faced significant "Out of Memory" (OOM) errors. With the model parameters occupying ~23.5 GB of VRAM, a batch size of 1 was initially the limit. **Solution:** We moved to the cluster and implemented Gradient Accumulation. We also researched LLaMA quantization (4-bit/8-bit) and Flash Attention, though full implementation of these is slated for future work.

5.2 Data Pipeline

Disk space was a critical issue. **Solution:** A rigorous cleanup script was written to remove '.tar' files immediately after extraction, freeing 250GB. We also implemented multi-threaded data loading to improve GPU utilization, which initially hovered below 30%.

6 Conclusion and Future Work

In this work, we successfully implemented "Bridge & Align," a Q-Former based architecture for Multilingual ASR. By training a lightweight interface between a frozen SeamlessM4T encoder and a LLaMA LLM, we achieved a stable alignment with a perplexity of 39.06. The transition from single-node to multi-node SLURM training reduced epoch time from 6 days to 12 hours, proving the scalability of the approach.

Future Work:

1. **Stage 2 Training:** Unfreezing the LLM (via LoRA) for translation and summarization tasks.
2. **Optimization:** Integrating Flash Attention and DeepSpeed to further reduce memory footprint.
3. **Inference:** rigorous WER calculation on the Kathbath test set.

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